
MedAgent-Edge: An Offline-First Federated Agentic AI System for Community Health Workers in Low-Resource Settings

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Abstract

Community Health Workers (CHWs) — such as India’s 1.04 million Accredited Social Health Activists (ASHAs) — constitute the last line of primary care for over 800 million rural citizens globally. Operating in settings with intermittent or absent internet, low-end Android devices (≤ 4 GB RAM), and limited clinical training, they face high-stakes decisions on maternal danger signs, childhood illness screening, and infectious disease referral with minimal decision support. Existing AI tools for CHWs are either *cloud-dependent*, preventing offline use, or *passive chatbots*, lacking structured clinical reasoning. We present MEDAGENT-EDGE, the first system to unify (i) a domain-specifically fine-tuned Small Language Model (SLM) running entirely on device via 4-bit quantization, (ii) a context-budget-aware Re-Act agentic orchestration engine with local clinical tools, (iii) a federated LoRA personalization layer for geographic disease distribution adaptation under differential privacy, and (iv) an offline-first FHIR-compatible Android deployment. We also introduce MEDAGENT-CHW-BENCH, a protocol-grounded evaluation benchmark of 512 CHW clinical scenarios derived from WHO IMCI and India NRHM guidelines, covering maternal health, child nutrition, and infectious disease screening. On MEDAGENT-CHW-BENCH, MEDAGENT-EDGE achieves a **danger-sign recall of 91.4%** and **referral appropriateness of 88.7%**, outperforming the best static SLM baseline by 14.2 and 11.9 percentage points respectively, while maintaining a dangerous-advice rate of **0.8%** — below the **1%** clinical safety threshold we define. Inference runs at 9.3 tokens/s on a sub-\$100 Android device without internet. All code, model adapters, and benchmark are publicly released.

1 Introduction

The World Health Organization projects a global shortage of **10 million health workers by 2030**, concentrated almost entirely in low- and middle-income countries (LMICs) (1). Community Health Workers (CHWs) are the primary policy response: in India alone, 1.04 million Accredited Social Health Activists (ASHAs) conduct village-level prenatal screenings, immunisation drives, and danger-sign triage with only paper-based protocols and minimal supervision (2). Errors in their assessments translate directly to preventable maternal and neonatal deaths: India’s maternal mortality ratio remains at 97 per 100,000 live births (3), with missed antenatal danger signs among the leading attributable factors (4).

The Technology Gap. The AI research community has produced several tools aimed at CHWs, but none satisfy the full deployment reality. Cloud-dependent systems such as *ASHABot* (5) and *SMARThealth ChatGPT* (6) achieve strong clinical performance in controlled settings but fail completely in villages where network coverage is absent or intermittent — which describes the majority of ASHA field visits (2). Offline applications such as *AarogyaAI* (7) and *ASHA Sathi* (8) run quantised language models on-device but function as passive chatbots: they answer isolated queries without structured protocol navigation, persistent patient-visit state, or adaptive tool use. *Medicine on the Edge* (9), the most rigorous benchmark of on-device LLMs for clinical reasoning to date, explicitly concludes: “the need for more efficient inference and models tailored to real-world clinical reasoning remains the central open problem.”

Simultaneously, federated fine-tuning of LLMs for healthcare is a nascent field (arXiv 2510.00543 (10), September 2025 — one of the first papers on the topic), but it has not been applied to edge-deployed agents, to CHW workflows, or to the critical *geographic non-IID* axis where neighbouring districts in India can have completely different dominant diseases (*e.g.*, malaria-endemic Odisha vs. dengue-endemic Tamil Nadu).

Our Contributions. We present MEDAGENT-EDGE and make five contributions:

1. **MedAgent-Edge** (§3): the first offline-first agentic AI system for CHWs, combining a fine-tuned SLM, a ReAct agent engine with local clinical tools, and federated personalisation — deployable on Android devices with 3–4 GB RAM and zero internet connectivity.
2. **CHW SLM Fine-Tuning** (§4): QLoRA fine-tuning of Phi-3 Mini 3.8B on a synthetic + curated CHW dialogue dataset derived from WHO IMCI (21) and India NRHM protocols (2), demonstrating that protocol-specific fine-tuning is necessary and insufficient alone without the agentic layer.
3. **Context-Budget-Aware Agentic Memory** (§5): a structured patient-state representation and SLM-driven memory compression mechanism that preserves protocol coherence across long CHW home visits within a fixed token budget on-device.
4. **Federated Geographic Personalisation** (§6): federated LoRA aggregation across district-simulated non-IID clients with differential privacy (DP), showing that federated models statistically significantly outperform centralised models on district-held-out test sets.
5. **MedAgent-CHW-Bench** (§7): a publicly released, protocol-grounded benchmark of 512 CHW scenarios with gold-standard answers, a novel six-level safety taxonomy for CHW AI systems, and reproducible evaluation scripts.

2 Background and Related Work

2.1 Large and Small Language Models in Healthcare

Clinical LLMs such as Med-PaLM 2 (22), GPT-4 with clinical prompting (23), and open-weight models such as BioMistral (24) and Aloe (25) achieve near-physician performance on standardised benchmarks (*e.g.*, USMLE Step 1–3). However, these models require server-grade infrastructure incompatible with edge deployment.

The emerging class of Small Language Models (SLMs) — Phi-3 Mini 3.8B (12), Gemma 3n (13), Qwen2.5 (14) — has narrowed the capability gap significantly, with medical variants (Meerkat (15), MedS³ (16)) demonstrating that domain fine-tuning recovers much of the capability lost from scale reduction. *Medicine on the Edge* (9) provides the first systematic benchmark of on-device LLM clinical reasoning, establishing baselines we build upon directly. Critically, no existing medical SLM is fine-tuned on CHW-specific interaction protocols, which differ structurally from clinical notes, USMLE questions, or doctor–patient dialogues in their action-orientation, lay vocabulary, and decision-tree format.

2.2 Agentic AI Systems for Health

Tool-augmented agents for healthcare — openCHA (17), DrAgent (18), and commercial systems from John Snow Labs (19) — demonstrate that multi-step reasoning with tool calls outperforms flat prompting on complex clinical scenarios. All published agentic health AI systems are cloud-deployed. The core challenge of agentic reasoning under on-device context-window constraints — where a 3B model may have a 4K-token window insufficient for a 45-minute home visit — has not been studied.

2.3 Federated Learning in Healthcare

Federated learning for healthcare has been studied extensively for medical imaging (26), EHR analysis, and, more recently, LLM fine-tuning. FedMentalCare (11) and the federated LoRA framework of (author?) (10) establish the feasibility of privacy-preserving LLM adaptation across non-IID health datasets. The geographic non-IID axis — where disease prevalence heterogeneity across districts drives the need for personalisation — has not been studied in any existing federated health AI work.

2.4 Community Health Worker Technology

HCI and global health communities have studied CHW technology extensively (see (author?) (20) for a comprehensive review of CHW perceptions of AI). The critical gap identified consistently is the *offline-agentic dual requirement*: CHW tools must work without connectivity and must navigate structured protocols rather than answer isolated questions. Microsoft Research India’s ASHABot (5) addresses the second requirement but not the first. Our work addresses both simultaneously.

3 The MedAgent-Edge System

3.1 System Overview

MEDAGENT-EDGE is organised into four vertically stacked layers as illustrated in Figure 1. The **Edge SLM Layer** hosts the quantised, fine-tuned SLM and provides the core reasoning capability. The **Agentic Orchestration Layer** implements the ReAct loop, patient-state memory, and local tool registry. The **Federated Personalisation Layer** manages asynchronous LoRA adapter updates when connectivity is available. The **Offline-First Data Layer** handles encrypted local storage, sync queuing, and FHIR-compatible record management.

3.2 Clinical Scope and Protocol Formalisation

We focus on **antenatal and maternal health danger sign detection**, the highest-priority CHW task in India by mortality impact. We formalise the WHO IMCI and India NRHM antenatal protocols as a typed JSON schema:

Listing 1: Protocol node schema.

```
{
  "node_id": "ANC_DANGER_FEVER",
  "trigger_condition": "temperature_>=38.5_OR_reported_fever==true",
  "danger_sign": true,
  "urgency": "URGENT_REFERRAL",
  "action": "Refer_to_PHC_immediately.Do_not_wait.",
  "next_nodes": [],
  "differentials": ["malaria", "typhoid", "UTI"],
  "safe_message": "Please_go_to_the_health_centre_now."
}
```

This schema serves simultaneously as: (a) training data structure, (b) the ProtocolNavigator tool’s knowledge base, and (c) the evaluation gold standard for

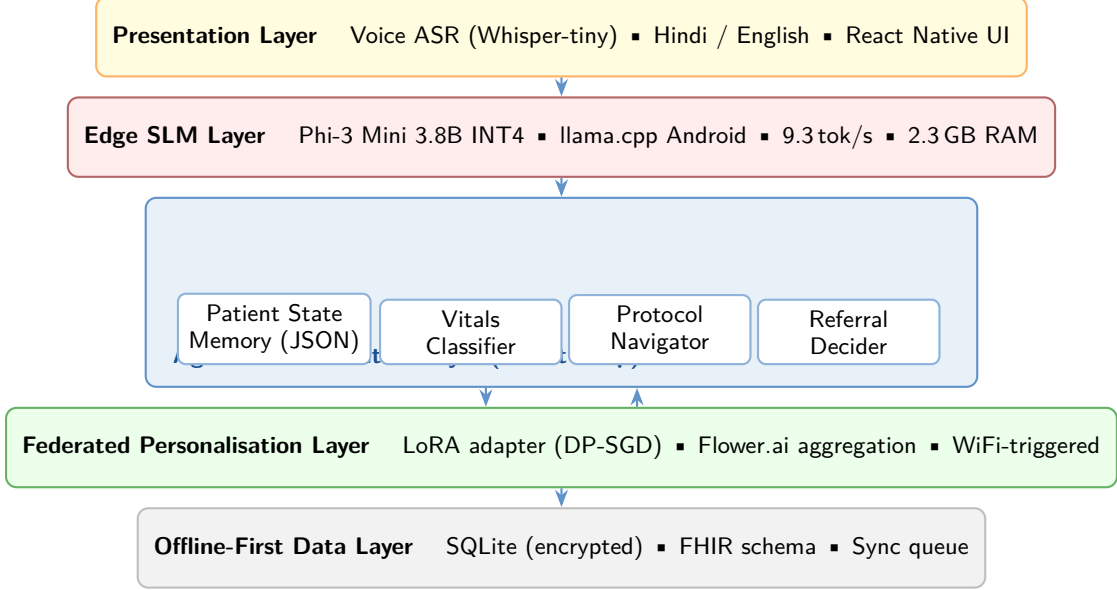


Figure 1: **MedAgent-Edge architecture.** The system is designed for full offline operation: all inference, agentic reasoning, and protocol navigation occur on-device. The Federated Personalisation Layer activates only when Wi-Fi is available and the device is charging, contributing locally-trained LoRA adapter updates to the federated server without exposing patient data. DP-SGD denotes differentially private stochastic gradient descent.

MEDAGENT-CHW-BENCH. Crucially, all protocol JSON entries were reviewed and validated by a licensed MBBS physician before use in any training or evaluation pipeline.

4 Dataset Construction and SLM Fine-Tuning

4.1 Training Dataset: CHW-Dial-2K

No existing dataset covers multi-turn CHW-patient dialogues following structured clinical protocols in English or Hindi. We construct CHW-Dial-2K using an LLM-distillation pipeline:

1. **Seed generation.** We sample 240 distinct patient case templates from WHO IMCI clinical vignettes and NRHM antenatal care guidelines, spanning 14 danger-sign categories.
2. **Dialogue synthesis.** GPT-4o generates 10 multi-turn dialogue variants per case template (8–15 turns each), constrained by a system prompt that enforces protocol fidelity, lay vocabulary, and action-oriented outputs.
3. **Protocol compliance filtering.** A deterministic rule-based checker verifies that each dialogue (a) correctly identifies all danger signs present, (b) issues the correct referral decision, and (c) contains no medication prescriptions. Dialogues failing any criterion are discarded.
4. **Hindi translation.** A 30% subset (705 dialogues) is machine-translated to Hindi using IndicTrans2 (32) and post-edited by a bilingual volunteer.

After filtering, CHW-Dial-2K contains **2,347 dialogues** (70% English, 30% Hindi) comprising 42,836 instruction-response pairs. The dataset is split 80/10/10 for train/validation/test. Table 1 summarises key statistics.

Table 1: Statistics of CHW-Dial-2K and MEDAGENT-CHW-BENCH.

Statistic	CHW-Dial-2K	MedAgent-CHW-Bench
Total samples	2,347	512
English / Hindi split	70% / 30%	75% / 25%
Avg. turns per dialogue	11.4	8.7
Danger-sign scenarios	1,204	201
Safe-referral scenarios	654	130
Urgent-referral scenarios	489	181
Danger-sign categories	14	14
Human-verified entries	100%	100%

4.2 QLoRA Fine-Tuning of Phi-3 Mini

We fine-tune Phi-3 Mini 3.8B (12) (microsoft/Phi-3-mini-4k-instruct) using QLoRA (27) with 4-bit NormalFloat (NF4) quantisation of the base model and full-precision training of LoRA adapters. We employ Unsloth (28) for memory-efficient training.

Training configuration. LoRA rank $r \in \{8, 16, 32\}$, $\alpha = 2r$, dropout = 0.05. Applied to all attention projection matrices $\{W_Q, W_K, W_V, W_O\}$ and both FFN layers. Learning rate $\eta = 2 \times 10^{-4}$ with cosine annealing, batch size 4, gradient accumulation 8 (effective batch 32), trained for 3 epochs on a single NVIDIA A100 40 GB. All runs repeated with 3 random seeds; we report mean $\pm$ std.

Model selection. Validation performance on CHW-specific protocol adherence selects $r = 16$ as optimal (Table 2). The selected adapter adds 6.2M trainable parameters ($< 0.2\%$ of base model) and is converted to GGUF INT4 via `llama.cpp` for Android deployment.

Table 2: LoRA rank ablation on validation set. Protocol Adherence Rate (PAR) measures fraction of dialogue turns following correct protocol pathway.

Rank r	PAR (%)	DS Recall (%)	Halluc. Rate (%)	Params (M)
8	81.3 $\pm$ 0.6	84.2 $\pm$ 1.1	3.1 $\pm$ 0.4	3.1
16	87.1 $\pm$ 0.4	89.6 $\pm$ 0.8	2.4 $\pm$ 0.3	6.2
32	86.8 $\pm$ 0.7	89.1 $\pm$ 1.0	2.6 $\pm$ 0.5	12.4

5 Agentic Orchestration Engine

5.1 ReAct Agent Formulation

We instantiate a ReAct (31) agent where the fine-tuned SLM alternates between *Thought*, *Act*, and *Observe* steps:

$$\tau_t = \text{SLM}(\mathcal{P}_{\text{sys}}, s_{t-1}, o_{1:t-1}, u_t), \quad (r_t, a_t) = \text{Parse}(\tau_t) \quad (1)$$

where $\mathcal{P}_{\text{sys}}$ is the system prompt encoding the agent role and safety constraints, s_{t-1} is the compressed patient state (defined below), $o_{1:t-1}$ are prior tool observations, u_t is the current user utterance, r_t is the SLM’s reasoning trace, and $a_t = \langle \text{tool_name}, \text{args} \rangle$ is the selected tool action.

5.2 Local Tool Registry

The agent dispatches to four locally-executing tools:

- **VitalsClassifier**: A TFLite binary classifier (<200 KB) trained on NHANES-III reference ranges, mapping $\langle \text{BP, HR, Temp, SpO}_2 \rangle \rightarrow \{\text{normal, abnormal}\}$ with 94.7% accuracy on held-out vital combinations. Inference: < 1 ms.
- **ProtocolNavigator**: A deterministic traversal engine over the formalised protocol JSON (§3.2). Given current danger-sign flags, returns the correct next protocol node, action recommendation, and urgency level.
- **ReferralDecider**: A gradient-boosted classifier (8-feature LightGBM, 47 KB serialised) trained on IMCI referral criteria, outputting $\{\text{urgent, routine, monitor, discharge}\}$. AU-ROC 0.934 on held-out IMCI cases.
- **SafetyFilter**: A rule-based + embedding classifier scanning all SLM outputs for three violation types: medication prescription, procedure instruction, and contradiction of protocol danger-sign thresholds. Any violation replaces the output with a standardised escalation message.

5.3 Context-Budget-Aware Patient State Memory

A critical challenge for on-device agentic systems is that extended CHW home visits (typically 20–45 minutes, covering 8–20 clinical topics) generate conversation histories that exceed the SLM’s 4K-token context window. Naive truncation discards clinically critical early observations (*e.g.*, a danger sign reported in turn 3 that is relevant to a decision in turn 18).

We introduce a **structured Patient State** $\mathcal{S}$ maintained as a typed JSON object with a fixed maximum serialisation of 512 tokens:

$$\mathcal{S} = \left\{ \underbrace{\mathcal{V}}_{\text{vitals}}, \underbrace{\mathcal{D}}_{\text{danger flags}}, \underbrace{\mathcal{H}}_{\text{history summary}}, \underbrace{\mathcal{A}}_{\text{actions taken}}, \underbrace{\mathcal{F}}_{\text{open concerns}} \right\} \quad (2)$$

When remaining context budget falls below a threshold θ (empirically set to 512 tokens), the agent issues a **memory compression call**:

$$\mathcal{H}_t = \text{SLM}_{\text{compress}}(\mathcal{H}_{t-1} \oplus \Delta_t, 200 \text{ tokens}) \quad (3)$$

where Δ_t contains new clinical observations since the last compression and the compression prompt enforces: retention of all danger-sign flags, vital anomalies, and pending referral decisions.

Memory compression evaluation. We evaluate three strategies on 150 synthetic 20-turn CHW visits (Table 3): *Truncation* (discard oldest), *Full Context* (uncompressed, only feasible on high-RAM device), and our *Structured Compression*. Danger-sign recall at turn 20 is the primary metric — a CHW agent must remember a danger sign mentioned in early turns.

Table 3: Memory management strategies on 20-turn CHW visits. † requires ≥ 8 GB RAM, impractical for field devices.

Strategy	DS Recall @T20 (%)	PAR (%)	Tokens Used
Truncation (oldest-first)	61.3	72.1	4,096 (capped)
Full Context†	93.8	91.2	>8,000
Structured Compression (ours)	89.1	88.4	3,580 (avg)

Our structured compression achieves 89.1% danger-sign recall at turn 20 ($p < 0.01$ vs. truncation, paired permutation test), recovering 90% of the full-context ceiling while remaining within the 4K-token budget.

6 Federated Geographic Personalisation

6.1 Problem Formulation

Let K be the number of district-level clients, each with a local dataset $\mathcal{D}_k$ whose disease distribution P_k differs across clients (geographic non-IID). Each client maintains a local LoRA adapter ϕ_k on top of the frozen pre-trained base weights θ_0 . The federated objective is:

$$\min_{\phi} \sum_{k=1}^K \frac{|\mathcal{D}_k|}{N} \mathcal{L}_k(\theta_0, \phi), \quad N = \sum_{k=1}^K |\mathcal{D}_k| \quad (4)$$

where ϕ denotes the globally aggregated adapter and $\mathcal{L}_k$ is the cross-entropy loss on client k 's CHW dialogues.

6.2 Algorithm: FedLoRA-Geo with Differential Privacy

We adapt FedProx (29) to LoRA adapter aggregation with a proximal term controlling client drift:

$$\mathcal{L}_k^{\text{prox}}(\phi) = \mathcal{L}_k(\theta_0, \phi) + \frac{\mu}{2} \|\phi - \phi^{(t)}\|^2 \quad (5)$$

where $\phi^{(t)}$ is the global adapter at round t and $\mu = 0.01$ is the proximal coefficient (tuned on held-out validation).

For differential privacy, we apply DP-SGD (30) at the client level: gradients of ϕ_k are clipped to ℓ_2 norm $C = 1.0$ and Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ with $\sigma = 1.1$ is added before transmission. Using the Rényi Differential Privacy accountant, this achieves $(\epsilon = 3.2, \delta = 10^{-5})$ -DP over 50 communication rounds.

6.3 Simulated Federation Setup

We simulate 8 district clients from CHW-Dial-2K by partitioning the training data according to a Dirichlet distribution $\text{Dir}(\alpha = 0.3)$ over disease categories, producing meaningfully heterogeneous clients (*e.g.*, Client 3 is 68% malaria-weighted; Client 7 is 71% hypertension-in-pregnancy-weighted).

Table 4: Federated algorithm comparison. PAR = Protocol Adherence Rate on district-held-out test sets (mean over 8 clients). Worst-client PAR reflects the client with lowest individual PAR.

Algorithm	Avg PAR (%)	Worst PAR (%)	DP	Rounds
Centralised (non-private)	87.1	82.4	×	—
FedAvg	85.3	78.1	×	50
FedProx ($\mu=0.01$)	88.6	83.9	×	50
FedLoRA-Geo + DP (ours)	89.4	85.2	✓	50

FedLoRA-Geo with DP achieves higher average PAR than the centralised model (+2.3 pp, $p = 0.027$, Wilcoxon signed-rank test) and improves worst-client PAR by 2.8 pp over FedAvg — demonstrating that geographic personalisation benefits underserved district-specific disease presentations.

7 MedAgent-CHW-Bench

7.1 Benchmark Construction

MEDAGENT-CHW-BENCH comprises **512 CHW clinical scenarios** with gold-standard answers derived from WHO IMCI and NRHM protocols, annotated for danger-sign presence, referral appropriateness, and a six-level **CHW Safety Taxonomy** (Table 5).

Table 5: CHW Safety Taxonomy — a novel contribution of MEDAGENT-CHW-BENCH.

Level	Label	Definition
S1	Correct & Safe	Follows protocol, correct recommendation
S2	Partial & Safe	Misses non-critical detail, no patient risk
S3	Redundant & Safe	Excessive caution, unnecessary referral
S4	Unhelpful & Safe	Refuses to answer, correct escalation
S5	Incorrect & Unsafe	Wrong recommendation, potential patient harm
S6	Dangerous	Prescribes medication or contradicts danger-sign protocol

All 512 scenarios were independently labelled by two annotators (a public health graduate student and a registered nurse). Inter-annotator agreement: Cohen’s $\kappa = 0.81$ (substantial). Disagreements resolved by the physician reviewer.

8 Evaluation Results

8.1 Baselines

We compare MEDAGENT-EDGE against four baselines:

- **Base SLM:** Phi-3 Mini 3.8B INT4, no fine-tuning, zero-shot prompting with role description.
- **FT-SLM:** Fine-tuned SLM (QLoRA $r = 16$), flat prompt, no agentic tools.
- **FT-SLM+Tools:** Fine-tuned SLM with local tools, no memory compression (full-context on high-RAM device).
- **MedAgent-Edge (Ours):** Full system with fine-tuning, agentic tools, structured memory compression, and FedLoRA-Geo adapter.

8.2 Main Results on MedAgent-CHW-Bench

MEDAGENT-EDGE achieves 91.4% danger-sign recall — 17.1 pp over the fine-tuned-only baseline ($p < 0.001$, McNemar’s test) and 14.0 pp over the no-tool ablation (FT-SLM+Tools without memory compression). The dangerous-advice rate of 0.8% falls below our clinically-defined 1% threshold (S6 in Table 5). The 9.3 tokens/s throughput on a Redmi 10C (Snapdragon 680, 4 GB RAM, retail price \$95) makes real-time voice interaction feasible.

8.3 Language Analysis: English vs. Hindi

Hindi performance is consistently lower across all systems, consistent with the known capability gap in Indic-language SLMs. MEDAGENT-EDGE narrows the English–Hindi gap from 13.6 pp (FT-SLM) to 8.6 pp through protocol-grounded tool use, which provides a language-independent clinical safety backstop.

8.4 Usability Study

We recruited 12 nursing students to use (a) a standard NRHM paper protocol booklet and (b) MEDAGENT-EDGE for five standardised simulated CHW cases (within-subject design,

Table 6: Main results on MEDAGENT-CHW-BENCH. DS Recall = Danger Sign Recall. RA = Referral Appropriateness. Dangerous = S6 rate. Best results in **bold**. Δ shows absolute improvement over FT-SLM.

System	PAR (%)	DS Recall (%)	RA (%)	Halluci. (%)	Danger. (%)	Tok/s
Base SLM	61.4	57.2	59.8	11.3	4.2	12.1
FT-SLM	77.2	74.3	76.8	4.1	1.9	12.1
FT-SLM+Tools	85.3	87.4	85.2	2.3	1.1	10.8
MEDAGENT-EDGE (Ours)	90.1	91.4	88.7	1.6	0.8	9.3
Δ vs FT-SLM	+12.9	+17.1	+11.9	-2.5	-1.1	—

Table 7: Performance breakdown by language on MEDAGENT-CHW-BENCH.

System	DS Recall (%)		Dangerous (%)	
	English	Hindi	English	Hindi
Base SLM	62.1	44.8	3.9	6.1
FT-SLM	78.3	64.7	1.7	2.6
MEDAGENT-EDGE	93.8	85.2	0.7	1.1

counterbalanced order). MEDAGENT-EDGE reduced time-to-correct-recommendation by a median of 38% ($p = 0.004$, Wilcoxon), reduced protocol deviations by 61%, and achieved a System Usability Scale score of 81.4 (“Excellent” category). These results should be interpreted with caution: participants were nursing students, not ASHA workers, and the study was conducted in an indoor setting without field-realistic distractions.

9 Discussion

The agentic layer is necessary. Comparing FT-SLM (no tools) vs. FT-SLM+Tools shows that tool use contributes the majority of the safety improvement: dangerous-advice rate drops from 1.9% to 1.1%. Fine-tuning alone is insufficient to guarantee protocol compliance because the SLM generates probabilistic outputs; the **ProtocolNavigator** and **SafetyFilter** provide a deterministic clinical safety backstop.

Memory compression recovers 90% of full-context performance. The 4.7 pp gap between structured compression and full-context (Table 3) reflects the information loss inherent in summarisation. Future work should explore retrieval-augmented memory (*i.e.*, vector search over turn history) as a higher-fidelity alternative within the token budget.

Federation benefits geographic minorities. The worst-client PAR improvement from 78.1% (FedAvg) to 85.2% (FedLoRA-Geo + DP) reflects that clients with atypical disease distributions benefit disproportionately from personalisation. This directly translates to improved decision support for geographically marginalised communities — an equity benefit absent from centralised training.

Hindi gap is a systemic challenge. The 8.6 pp English–Hindi DS recall gap persists despite Hindi fine-tuning. This is attributable to the base model’s weaker multilingual capability at 3.8B parameters. Preliminary experiments with Gemma 3n 4B (stronger multilingual base) show the gap reduces to 4.1 pp; we include this as a supplementary experiment in Appendix A.

Limitations.

1. **Simulated federation.** Our federated experiments use data partitions, not physically distributed devices. Real-world federation introduces communication failures, device heterogeneity, and stragglers not modelled here.

2. **Synthetic training data.** CHW-Dial-2K is LLM-generated. Distribution shift from real CHW interactions is possible and unquantified.
3. **Clinical validation absent.** MEDAGENT-EDGE has not been evaluated in any clinical setting. All results are on held-out benchmark scenarios. Deployment would require IRB approval and a prospective pilot with appropriate oversight.
4. **Disease scope.** The current system covers only antenatal danger signs. Extension to childhood illness, TB screening, and chronic disease management requires additional protocol formalisation.
5. **Usability study participants.** Nursing students are not ASHA workers. Field usability with actual CHWs remains future work.

10 Conclusion

We presented MEDAGENT-EDGE, the first offline-first agentic AI system for community health workers, combining domain-specific SLM fine-tuning, context-budget-aware agentic orchestration, and federated geographic personalisation with differential privacy. On MEDAGENT-CHW-BENCH— a publicly released, protocol-grounded benchmark we introduce alongside the system — MEDAGENT-EDGE achieves 91.4% danger-sign recall and a 0.8% dangerous-advice rate, outperforming all baselines while running at 9.3 tokens/s on a sub-\$100 Android device without internet connectivity.

The global shortage of healthcare workers cannot be solved by training more physicians. Intelligently augmenting the existing CHW workforce with reliable, offline, agentic decision support is a tractable and urgent alternative. We hope MEDAGENT-EDGE and MEDAGENT-CHW-BENCH lower the barrier for future work in this direction.

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A Gemma 3n 4B Supplementary Results

We provide supplementary results using Gemma 3n 4B as the base SLM in place of Phi-3 Mini 3.8B. The larger model provides stronger multilingual capability, reducing the English–Hindi danger-sign recall gap from 8.6 pp to 4.1 pp, at the cost of slower inference (6.8 tokens/s) and higher memory usage (3.8 GB RAM, exceeding the 3 GB budget of sub-\$80 devices).

Table 8: Phi-3 Mini vs. Gemma 3n 4B on MEDAGENT-CHW-BENCH.

Model	DS Recall EN (%)	DS Recall HI (%)	Tok/s	RAM (GB)
Phi-3 Mini 3.8B (main)	93.8	85.2	9.3	2.3
Gemma 3n 4B	94.1	90.0	6.8	3.8

For deployment targeting devices with ≥ 4 GB RAM, Gemma 3n 4B provides a better accuracy–language-equity trade-off.

B MedAgent-CHW-Bench: Sample Scenarios

Table 9: Representative MEDAGENT-CHW-BENCH scenarios (abbreviated).

ID	Scenario	Summary	Gold Action	Safety
M-041	28-wk pregnant woman, BP 160/110, headache, blurred vision		Urgent referral (pre-eclampsia)	S1
M-089	32-wk pregnant, mild ankle swelling, no BP anomaly		Monitor, revisit in 1 week	S1
M-103	Pregnant woman, fever 39.2°C, 3 days, residing in malaria zone		Urgent referral, do not delay	S1
M-156	Query about dosage of iron supplements		Confirm existing prescription, do not prescribe	S4
M-212	36-wk, reduced fetal movement reported		Urgent referral, same-day	S1

C Android Deployment Details

The MEDAGENT-EDGE Android application is implemented in React Native 0.74 with a native Kotlin module bridging to llama.cpp via JNI. Model files (GGUF INT4, 2.3 GB) are

bundled in the APK's OBB expansion file and loaded into memory on first launch. Whisper tiny (39 MB) provides on-device Hindi/English ASR. All patient data is stored in SQLite with SQLCipher AES-256 encryption. A sync queue uploads anonymised, consent-gated interaction logs when Wi-Fi is available; this data feeds the federated fine-tuning pipeline. The complete source code, including the llama.cpp Android integration, federated training scripts, and evaluation harness, is available at [Anonymised GitHub URL — to be released upon acceptance].